the past can bear the imprint of today's actions. If we insert photon detectors along the two pathways light takes to a screen, then our story of the past will include a description of which pathway each photon took; by inserting the photon detectors, we ensure that which-path information is an essential and definitive detail of our story. But, if we don't insert the photon detectors, our story of the past will, of necessity, be different. Without the photon detectors, we can't recount anything about which path the photons took; without the photon detectors, which-path details are fundamentally unavailable. Both stories are valid. Both stories are interesting. They just describe different situations.

An observation today can therefore help complete the story we tell of a process that began yesterday, or the day before, or perhaps a billion years earlier. An observation today can delineate the kinds of details we can and must include in today's recounting of the past.

Erasing the Past

It is essential to note that in these experiments the past is not in any way altered by today's actions, and that no clever modification of the experiments will accomplish that slippery goal. This raises the question: If you can't change something that has already happened, can you do the next best thing and erase its *impact* on the present? To one degree or another, sometimes this fantasy can be realized. A baseball player who, with two outs in the bottom of the ninth inning, drops a routine fly ball, allowing the opposing team to close within one run, can undo the impact of his error by a spectacular diving catch on the ball hit by the next batter. And, of course, such an example is not the slightest bit mysterious. Only when an event in the past seems definitively to preclude another event's happening in the future (as the dropped fly ball definitively precluded a perfect game) would we think there was something awry if we were subsequently told that the precluded event had actually happened. The *quantum eraser,* first suggested in 1982 by Marlan Scully and Kai Drühl, hints at this kind of strangeness in quantum mechanics.

A simple version of the quantum eraser experiment makes use of the double-slit setup, modified in the following way. A tagging device is